

CL 326: Integrated Chemical Engineering Laboratory

Heat Transfer in an Agitated Vessel

Experiment 6

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Submitted by

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1 Aim

To determine the overall heat transfer coefficient, based on the outside area of the cooling coil, for an agitated vessel operating under unsteady-state batch-cooling conditions.

2 Apparatus

- Cylindrical vessel fitted with an immersed cooling coil.
- Variable-speed agitator and speed regulator.
- Electric heater with temperature controller.
- Cooling-water circulation pump, reservoir and connecting lines.
- Digital indicators for vessel, cooling-water inlet and cooling-water outlet temperatures.
- Stopwatch for recording the batch-cooling time.

3 Experimental Setup

The hot test liquid was held in an agitated vessel and cooling water was circulated through the immersed coil. Agitation reduced temperature gradients within the vessel, allowing the measured vessel temperature to be treated as the instantaneous bulk-liquid temperature.



Figure 1: Experimental setup used for heat transfer in the agitated vessel.

4 Procedure

1. The vessel was filled with the test liquid until the cooling coil was completely immersed. The mass of liquid charged was noted.

2. The agitator was started and adjusted to the selected operating speed.
3. The cooling-water pump was connected to the supply and started. Initially, the water flow was kept low to avoid excessive heating of the coil.
4. The electric heater was switched on and the controller was set to heat the vessel contents to approximately 100 °C.
5. Near the required temperature, the cooling-water flow was increased to its maximum operating value.
6. On reaching 100 °C, the heater was switched off and the stopwatch was started simultaneously.
7. The elapsed time was recorded for every 5 °C decrease in vessel temperature. The cooling-water inlet and outlet temperatures were recorded at each point, down to a final vessel temperature of 50 °C.

5 Theory

Batch heating and cooling are unsteady-state operations because the temperature driving force, and hence the heat-transfer rate, changes continuously with time. Such operations are used when a fixed quantity of material must be heated or cooled for a specified period, as in batch reactors, mixing vessels, food-processing tanks and pharmaceutical equipment.

Consider an agitated vessel containing W kg of a hot liquid having specific heat C_p . The liquid is at a uniform bulk temperature T because of mixing. Cooling water flows through an immersed coil at a mass flow rate m , has specific heat c_p , and enters and leaves at temperatures t_1 and t_2 , respectively. For the cooling water,

$$mc_p(t_2 - t_1) = UA \left[\frac{t_2 - t_1}{\ln\left(\frac{T - t_1}{T - t_2}\right)} \right], \quad (1)$$

where U is the overall heat transfer coefficient based on the outside coil area A . Cancelling the common temperature-rise term gives

$$\ln\left(\frac{T - t_1}{T - t_2}\right) = \frac{UA}{mc_p}. \quad (2)$$

Defining the temperature ratio k ,

$$k = \frac{T - t_1}{T - t_2} = \exp\left(\frac{UA}{mc_p}\right), \quad (3)$$

and therefore

$$t_2 = T - \frac{T - t_1}{k}. \quad (4)$$

An unsteady-state energy balance on the well-mixed vessel gives

$$-WC_p \frac{dT}{d\theta} = mc_p(t_2 - t_1) = mc_p(T - t_1) \left(\frac{k-1}{k} \right). \quad (5)$$

Integrating from $T = T_1$ at $\theta = 0$ to $T = T_2$ at elapsed time θ yields

$$\theta = \frac{WC_p}{mc_p} \ln \left(\frac{T_1 - t_1}{T_2 - t_1} \right) \left(\frac{k}{k-1} \right). \quad (6)$$

Equation (6) is used to determine k from the complete cooling run. Once k is known, the overall coefficient is obtained from

$$U = \frac{mc_p \ln k}{A}. \quad (7)$$

6 Observations

The measured readings and the point-wise quantities calculated from the Excel sheet are listed in Table 1.

Table 1: Observed batch-cooling data and point-wise calculations.

Sr. No.	T (°C)	θ (s)	θ (h)	t_1 (°C)	t_2 (°C)	K_i	U_i (kcal h ⁻¹ m ⁻² °C ⁻¹)
1	100	0	0.0000	22.9	26.1	1.0433	332.2
2	95	81	0.0225	22.6	25.8	1.0462	354.3
3	90	161	0.0447	22.6	25.7	1.0482	369.0
4	85	244	0.0678	22.6	25.5	1.0487	373.0
5	80	333	0.0925	22.6	25.4	1.0513	391.9
6	75	433	0.1203	22.6	25.3	1.0543	414.6
7	70	554	0.1539	22.6	25.2	1.0580	442.1
8	65	689	0.1914	22.9	25.2	1.0578	440.3
9	60	845	0.2347	23.1	25.2	1.0603	459.2
10	55	1026	0.2850	23.4	25.2	1.0604	459.6
11	50	1252	0.3478	23.4	25.1	1.0683	517.6
Average point-wise temperature ratio, $\bar{K}$						1.0543	—

Experimental constants

$$W = 6.9 \text{ kg},$$

$$m = 500 \text{ kg h}^{-1},$$

$$A = 0.0638 \text{ m}^2.$$

$$C_p = 0.635 \text{ kcal kg}^{-1} \text{ °C}^{-1},$$

$$c_p = 1.0 \text{ kcal kg}^{-1} \text{ °C}^{-1},$$

7 Calculations

7.1 Method 1: Point-wise temperature-ratio method

For every observation,

$$K_i = \frac{T_i - t_{1,i}}{T_i - t_{2,i}}, \quad U_i = \frac{mc_p \ln K_i}{A}. \quad (8)$$

For the first reading,

$$K_1 = \frac{100 - 22.9}{100 - 26.1} = 1.04330,$$

$$U_1 = \frac{500(1.0) \ln(1.04330)}{0.0638} = 332.2 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^\circ\text{C}^{-1}.$$

The Excel average of the eleven ratios is $\bar{K} = 1.054268$. Therefore,

$$U_{M1} = \frac{500(1.0) \ln(1.054268)}{0.0638} = 414.2 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^\circ\text{C}^{-1}. \quad (9)$$

7.2 Method 2: Integrated batch-cooling method

The measured time for cooling from $T_1 = 100^\circ\text{C}$ to $T_2 = 50^\circ\text{C}$ was

$$\theta = 1252 \text{ s} = \frac{1252}{3600} = 0.34778 \text{ h}.$$

Following the Excel calculation, the initial cooling-water inlet temperature $t_1 = 22.9^\circ\text{C}$ was used in the integrated equation. Define

$$\begin{aligned} \alpha &= \frac{WC_p}{mc_p} \ln \left(\frac{T_1 - t_1}{T_2 - t_1} \right) \\ &= \frac{6.9(0.635)}{500(1.0)} \ln \left(\frac{100 - 22.9}{50 - 22.9} \right) \\ &= 0.0091623 \text{ h}. \end{aligned}$$

Hence, from Eq. (6),

$$\begin{aligned} \frac{k}{k-1} &= \frac{\theta}{\alpha} = \frac{0.34778}{0.0091623} = 37.9574, \\ k &= \frac{37.9574}{37.9574 - 1} = 1.02706. \end{aligned}$$

Using Eq. (7),

$$\begin{aligned} U_{M2} &= \frac{500(1.0) \ln(1.02706)}{0.0638} \\ &= 209.2 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ }^\circ\text{C}^{-1} \\ &\approx 243.3 \text{ W m}^{-2} \text{ K}^{-1}. \end{aligned}$$

8 Results and Discussion

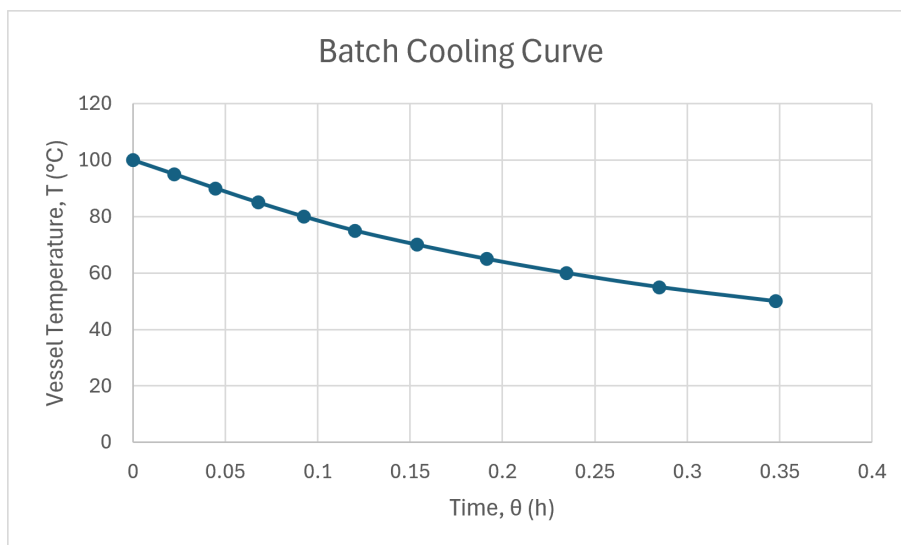


Figure 2: Vessel temperature plotted against elapsed cooling time.

Table 2: Overall heat transfer coefficient obtained by the two data-reduction methods.

Method	Ratio used	U (kcal h ⁻¹ m ⁻² °C ⁻¹)
Point-wise ratio average	$\bar{K} = 1.0543$	414.2
Integrated batch balance	$k = 1.0271$	209.2

The vessel cooled from 100 °C to 50 °C in 1252 s. The early 5 °C drops required about 80–90 s, whereas the final 5 °C drop required 226 s. Thus, the cooling curve becomes progressively less steep as the temperature difference between the vessel and the cooling water decreases.

The cooling-water temperature rise across the coil also decreased from 3.2 °C at the start to 1.7 °C at the final reading. The point-wise calculation is highly sensitive to this small temperature rise: errors of only a few tenths of a degree directly affect K_i and are magnified through $\ln K_i$. This is reflected in the increase of the calculated U_i values from 332.2 to 517.6 kcal h⁻¹ m⁻² °C⁻¹.

The point-wise estimate is nearly twice the integrated value. Since the integrated method uses the complete cooling time and follows the unsteady-state energy balance from which the working equation was derived, it is less dependent on any single inlet or outlet temperature reading. It is therefore selected as the representative result for this run:

$$U = 209.2 \text{ kcal h}^{-1} \text{ m}^{-2} \text{ °C}^{-1} \quad (243.3 \text{ W m}^{-2} \text{ K}^{-1})$$

9 Assumptions

- The vessel contents were perfectly mixed, so the bulk-liquid temperature was spatially uniform at each instant.
- Heat loss from the vessel to the surroundings was negligible.
- The specific heats and other physical properties of both fluids remained constant over the measured temperature range.

- The cooling-water mass flow rate remained at 500 kg h^{-1} .
- The initial inlet temperature, 22.9°C , represented the inlet condition in the integrated model; the small measured variation was neglected.
- Heat stored in the vessel wall, coil and agitator was negligible compared with that stored in the liquid charge.
- Radiation and phase change were absent.

10 Applications

- Heating and cooling of batch reactors used for neutralisation, polymerisation and crystallisation.
- Temperature control in pharmaceutical formulation and biochemical reactors.
- Batch pasteurisation, beverage blending and other food-processing operations.
- Estimation of heating and cooling times during scale-up of agitated vessels.
- Design and performance evaluation of internal coils and jackets used for process utilities.

11 Conclusion

The overall heat transfer coefficient for the agitated-vessel cooling coil was determined from the experimental batch-cooling data. The point-wise ratio method gave $414.2 \text{ kcal h}^{-1} \text{ m}^{-2} ^\circ\text{C}^{-1}$, while integration of the full unsteady-state balance gave $209.2 \text{ kcal h}^{-1} \text{ m}^{-2} ^\circ\text{C}^{-1}$. The integrated value was taken as the final result because it uses the complete cooling history and is less sensitive to the small measured temperature rise of the cooling water. The observed reduction in cooling rate with time was consistent with the decreasing thermal driving force.

12 References

1. CL 326 laboratory manual, "Heat Transfer in Agitated Vessel."
2. Excel readings recorded on 25 August 2026 during experiment.

13 Acknowledgement

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